

Rishuv Gorka

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EDUCATION

Indian Institute of Technology, Bombay
B.Tech., Electrical Engineering

Aug'2019 - Dec'2023

WORK EXPERIENCE

Software Developer Intern | Ensurezy

(Aug'2025 - Nov'2025)

- Designed and developed an end-to-end motor insurance policy issuance journey including quote generation, KYC verification, and payment integration
- Implemented the industry's first instant payout calculation and payment for agents by integrating Easebuzz services
- Worked in a team of 4 members to build scalable policy issuance workflows for 10+ insurers
- Technologies & Tools used: Python, Django, PostgreSQL, Celery, Redis, AWS, JavaScript, CI/CD

Project Research Assistant | IIT Bombay

(Jan'2024 - May'2025)

- Developed an optimization tool that automates crew scheduling for DMRC, using Integer Programming (Set Partitioning Problem) for dual-point scheduling and a Pyomo-based optimization model that reduces manual time by 90% while ensuring 100% compliance with operational constraints.
- Utilized Python, Pyomo, Pandas, Minotaur solver and scheduling algorithms to get the optimal solution, resulting in 12% reduction in total duties and with zero manual intervention and its solution has been in real-time use by DMRC since Feb. 2025.
- In addition, collaborated on the Time-tabling and line planning project for DMRC and Crew scheduling for Mumbai suburban rail, contributing to algorithm design, code reviewing and logic debugging, ensuring alignment with real-world railway constraints.
- Gained hands-on experience with Bash scripting, Ubuntu, and Git, while working on computational servers and solvers; also managed an AWS EC2 server for the deployment and maintenance of the crew scheduling tool.

KEY PROJECTS

Multi-Asset Portfolio Analysis | (Self Project)

(May'2025)

- Build a high impact portfolio analysis tool using 10+ years of data across equities, bonds, commodities, and crypto.
- Achieved 50%+ improvement in Sharpe ratio through mean-variance optimization and Monte Carlo simulations.
- Used Python, yFinance, Pandas to provide detailed insights through advanced visualizations, including drawdowns, rolling volatilities, and the efficient frontier.

Multi-Timeframe Trading | (Self Project)

(Jun'2025)

- Designed and implemented a multi-timeframe trading strategy in Python using EMA/SMA pullbacks and RSI for confirmation, with 15m entry and 1h confirmation logic for improved accuracy.
- Implement a backtesting system using backtesting.py to test the strategy with realistic order execution and stop loss/take profit handling, and export trade logs for performance analysis.
- Integrated Binance Testnet APIs for live deployment, enabling secure trading automation with real-time signal handling, persistent position tracking, and a crash-resilient logging system built with pandas and Python-binance.

MyNotes | (Self Project)

(Jun'2023-Sep'2023)

- Developed a web-based note-making application using Python, Django, HTML, and Bootstrap, allowing users to efficiently organize and manage their notes.
- Implemented secure user authentication with Django, incorporating features such as Sign Up, Login, and Logout for personalized access.
- Designed robust CRUD functionality, allowing users to create, edit, and manage notes, with added support for images, audio files, and automatic date-timestamping.

TECHNICAL SKILLS

Languages: Python, C++, Bash, SQL, HTML, CSS

Libraries: Django, Pandas, Matplotlib, Plotly, Pyomo

Softwares & others: Git, VS Code, Linux, Cloud Servers (AWS)

Solvers: Gurobi, Minotaur

SCHOLASTIC ACHIEVEMENTS

- Secured 99.3% percentile in JEE-Mains. (2019)
- Ranked among the top 50 candidates nationwide in the NDST exam, chosen from over 40,000 applicants. (2017)

POSITION OF RESPONSIBILITY

Mentor | Dakshana India

(2020 & 2021)

- This program aims to guide students who have qualified for JEE who were part of the Dakshana Foundation.
- Mentored students who have recently qualified JEE to help them choose their branches and colleges wisely.
- Helped to make a smooth transition into respective colleges and obtain financial help if needed.

EXTRA-CURRICULAR ACTIVITIES

- Visited Govt. schools in Mumbai to enlighten students of 10th and 12th class about Dakshana Selection Test. (2020)
- Completed NSO basketball course; secured top spots in Hostel-6 Sports GC & Aavhan Sports League. (2023)
- Fabricated a uniquely designed working Remote Controlled Plane for RC plane competition (2019)